Model Development Document: Mortgage CECL & DFAST Credit Loss Engine

> **Status:** Draft for independent validation. **As-of date:** 202512. **Data:** Freddie Mac Single-Family Loan-Level Dataset, sample files, Release 48. Every figure in this document is rendered from `results/metrics.json` by `scripts/build_reports.py`; none is typed by hand. Design decisions are referenced as D-xxx (Appendix A, full text in `DECISIONS.md`).

Executive summary

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Executive summary

This document describes a loan-level credit loss engine for US first-lien single-family mortgages, used for two purposes: the **CECL allowance** (ASC 326) and **DFAST-style supervisory stress projections**. Both uses share one set of models:

- a **12-month PD scorecard** (WoE logistic regression) used for ranking and monitoring;
- a **lifetime competing-risk hazard** for monthly default and prepayment, driven by the Federal Reserve's own scenario variables;
- a two-part **LGD** model (probability of cure × fractional-logit severity) and an **EAD** equal to the scheduled amortization balance;
- a monthly **projection engine** combining them to loan maturity.

Key results.

- **CECL:** the probability-weighted allowance is **0.70% of UPB** on 347,613 loans. The model-only allowance (without reversion to historical loss rates) is 0.34%.
- **DFAST:** under the 2026 severely adverse scenario, the 9-quarter loss rate is **1.02%**, against 0.21% under the baseline. The Federal Reserve's all-bank first-lien figure is 1.5%.
- **Validation evidence:**
    - out-of-time scorecard Gini 0.778
    - out-of-time prepayment actual/predicted 1.02
    - a replay of the 2008–2010 crisis reproducing realized losses (1.88% predicted vs 1.88% actual)
    - an untouched Jan–Jun 2026 backtest

Most material judgments:

- the historical loss window used for CECL reversion (D-035), which moves the allowance more than any other parameter;
- the post-2013 servicing-regime assumption in LGD (D-031, D-040);
- the exclusion of COVID policy months from hazard estimation (D-024).

Known limitations are in Section 9.

1. Purpose, scope and use

Intended use. Estimate lifetime expected credit losses for the CECL allowance and project 9-quarter credit losses under supervisory scenarios, for a portfolio of conforming fixed- and adjustable-rate first-lien mortgages.

In scope: loan-level PD, prepayment, LGD, EAD; scenario-conditional projection; allowance and stress-loss aggregation.

Out of scope:

- jumbo, subprime, HELOC and junior liens;
- pre-provision net revenue, capital ratios and balance-sheet dynamics;
- quantified qualitative adjustments (Q-factors);
- independent validation, which a model developer cannot perform on their own model.

Model risk tier (proposed). High materiality: the model drives a financial-statement estimate and a regulatory exercise (SR 26-2 materiality: exposure and purpose).

2. Data

2.1 Sources

Source	Content	Use
Freddie Mac SFLLD, sample files (Release 48)	1,374,984 loans, monthly performance through 202606, actual loss fields	All models
Federal Reserve 2026 DFAST files	Historic and scenario paths (unemployment, mortgage rate, Z.1 house prices)	Hazard drivers, CECL and DFAST scenarios
Freddie Mac House Price Index (FMHPI)	Monthly state and national indices	Mark-to-market LTV; state betas
FRED	30-year mortgage rate (PMMS), unemployment, recession dates	Refinance incentive, checks

2.2 Processing and data quality

Raw pipe-delimited files are read as text and cast with strict typing against the Release 48 layout (D-002), so layout drift fails rather than producing silent nulls. Conversion took 60 seconds with 1.0 GB peak memory (8.8 GB of text to 0.69 GB of parquet). Checks performed:

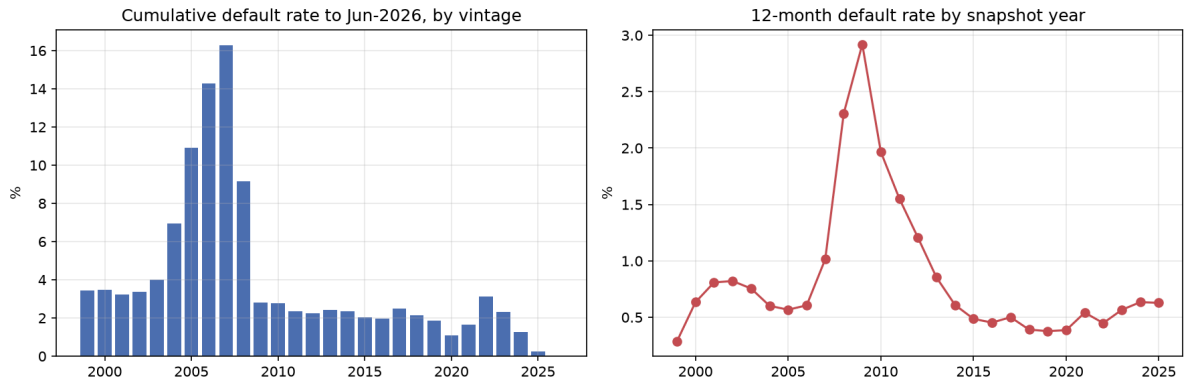
- 0 duplicate loan-months;
- 202 loans with month gaps (1,122 missing loan-months), kept, because events are keyed on period;
- 0 unknown zero-balance codes;
- the loss identity (UPB + accrued interest + expenses + recoveries = actual loss, recoveries stored as negatives) holds for 19,681 of 19,681 liquidations with loss data.

2.3 Target definitions

- **Default** (D-003): the first month with 90+ days past due, REO acquisition, or a default-type zero-balance code (02/03/09/15). Default is absorbing.
- **Forbearance** (D-004): 90+ DPD months during forbearance, payment deferral or disaster relief do not trigger default. Without this, 12-month default rates for snapshots taken in 2019 and 2020 (2.03%, 2.39%) would approach the 2009 peak (2.92%) despite negligible losses. With it, they are 0.38% and 0.39%. The flags exist only from 2014.
- **Prepayment** (D-005): voluntary payoff with scheduled term remaining.
- **Censoring** (D-006): reperforming-loan sales, defects, maturity, and loans active at the cutoff.

Vintage	Loans	Lifetime default (used)	Lifetime default (strict 90+ DPD)	12m snapshot PD
1999	50,000	3.45%	3.46%	0.81%
2002	50,000	3.36%	3.40%	0.78%

Vintage	Loans	Lifetime default (used)	Lifetime default (strict 90+ DPD)	12m snapshot PD
2005	50,000	10.90%	11.04%	1.73%
2007	50,000	16.29%	16.53%	3.52%
2008	50,000	9.16%	9.34%	2.25%
2011	50,000	2.35%	3.10%	0.37%
2014	50,000	2.36%	3.89%	0.40%
2017	50,000	2.51%	5.85%	0.52%
2020	50,000	1.10%	2.31%	0.25%
2023	50,000	2.33%	2.62%	0.83%
2026	24,988	0.02%	0.02%	0.00%



Default rates by vintage

3. 12-month PD scorecard

Method. The scorecard is fitted on yearly loan snapshots (D-007).

- Monotonic WoE binning (D-014), with minimum-bin exceptions for rare high-risk states (D-018).
- Selection: IV ≥ 0.02, a correlation filter, and backward elimination on wrong signs and on p-values using loan-clustered standard errors (D-015).
- Era-proxy features are excluded on training-only evidence (D-019).
- Samples: fit on 80% of loans from 1999–2014, an in-time holdout on the other 20%, and an out-of-time test on 2015–2019 vintages (D-012).

Sample	Snapshots	Defaults	Scorecard Gini	Scorecard KS	LightGBM Gini	Actual/predicted (scorecard)
fit	3,410,091	35,753	0.808	0.646	0.854	1.00
holdout	856,169	8,686	0.812	0.654	0.831	0.96
oot	1,151,976	5,082	0.778	0.632	0.791	0.54

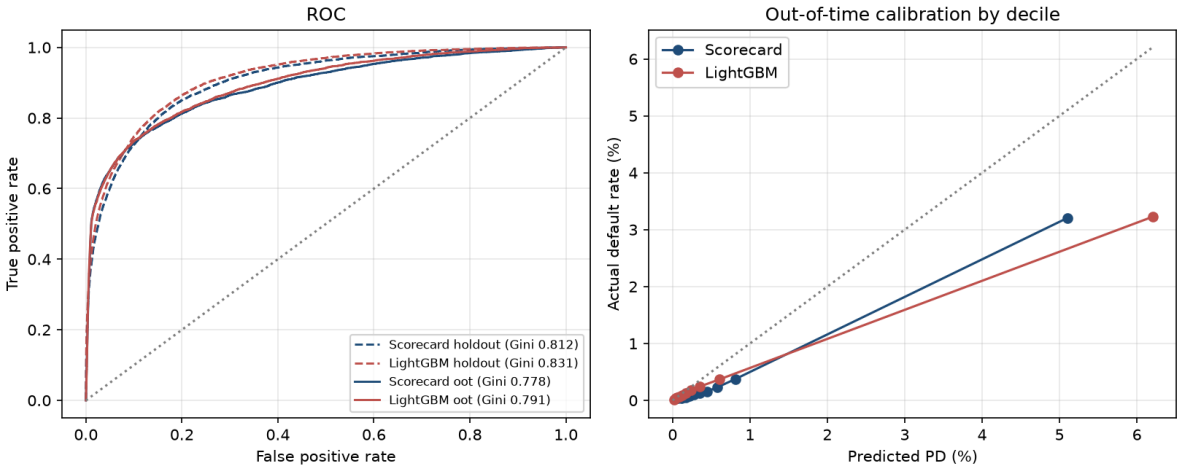
The out-of-time Gini's 95% loan-clustered bootstrap interval is 0.767–0.788. Score PSI between development and out-of-time is 0.072.

Robustness:

Check	Out-of-time Gini
Calendar split (training snapshots ≤ 2013, testing ≥ 2015)	0.772
Origination-only features	0.483
Loans current at the snapshot	0.573

Challenger. A monotone-constrained LightGBM adds +0.013 Gini (D-016). The scorecard is retained for transparency and stability.

Calibration. Out-of-time actual/predicted is 0.54: the scorecard reflects through-the-cycle risk from a crisis-inclusive training period. A leakage-free recalibration test (fit on 2015–2017, test on 2019 onward) is documented (D-020). Point-in-time PD levels for CECL come from the macro-driven hazard, not the scorecard.



ROC and calibration

4. Lifetime default and prepayment hazard

Specification (D-021, D-022). A monthly multinomial logit models survive, default and prepay as competing risks, giving $S(t) = \Pi(1 - h_d - h_p)$.

- **Drivers:** loan age (piecewise linear), origination risk (FICO, CLTV, DTI, rate spread, loan size, purpose, occupancy, single borrower), and three scenario variables:
  - unemployment (level and 4-quarter change, Fed definition);
  - the refinance incentive against the mortgage rate;
  - mark-to-market LTV from state house prices.
- **Estimation:** choice-based sampling with corrected constants (D-023). COVID policy months are excluded from estimation (D-024), and there is no forbearance covariate (D-025).

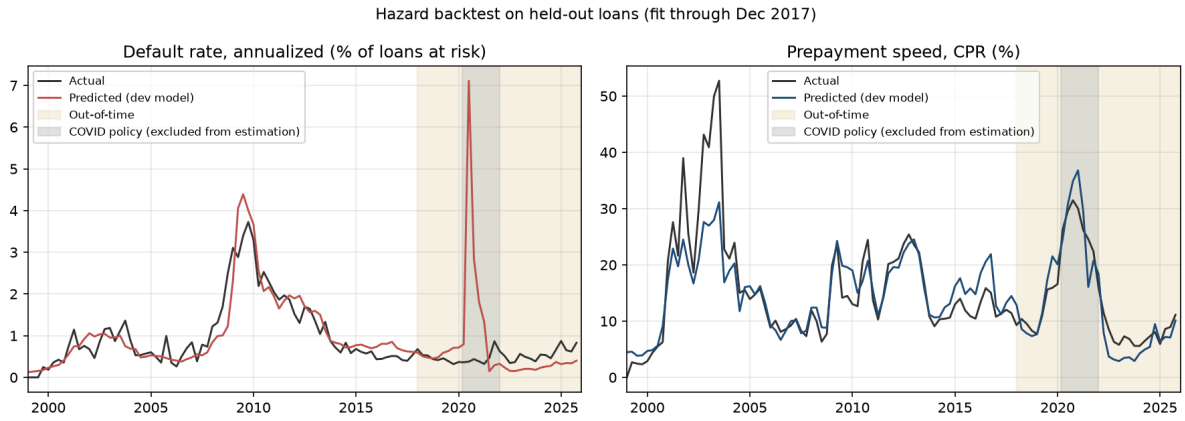
Backtest. The development model is fitted through 2017 and tested on held-out loans.

- **In-time:** default actual/predicted 0.97, prepay 1.01.
- **Out-of-time, 2018–2025:** prepayment actual/predicted is 1.02. Default is 0.69 overall:
  - COVID policy months come in at 0.22;
  - loans with forbearance history produce a large share of later defaults (handled as an overlay, D-026);
  - for loans never in forbearance, post-COVID actual/predicted is 1.08, 1.14 and 1.08 by vintage group.

Year	Defaults actual	Defaults predicted	Default A/P	Prepay A/P
2018	160	149	1.07	1.02
2019	124	189	0.65	0.85
2020	116	848	0.14	0.93
2021	131	237	0.55	0.98
2022	117	55	2.13	1.27
2023	132	54	2.45	2.02
2024	171	88	1.93	1.14
2025	247	116	2.12	1.13

Stress sensitivities (production model):

Shock	Default multiplier	Prepay multiplier
Unemployment +2 pts	1.56	1.12
House prices –20%	1.54	0.76
Mortgage rate –1 pt	1.19	1.71



Hazard backtest

5. Loss given default and exposure at default

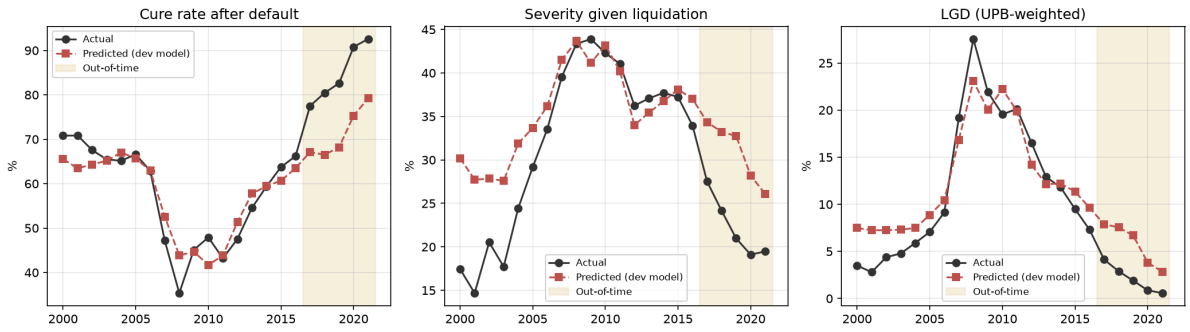
Structure (D-027). E[LGD] = (1 − P(cure)) × E[severity | liquidation], developed on resolved defaults through 2021.

- **Severity:** measured on principal, with UPB at default as the denominator, undiscounted (D-029).
- **Drivers:** mark-to-market LTV at default, the forward 12-month house-price path (D-030), real (HPI-deflated) loan size (D-041), MI coverage, foreclosure timeline, and property and occupancy type.
- **Cure model:** adds a post-2013 servicing-regime indicator; unemployment is excluded (D-031).

Results.

- **In-sample (defaults ≤ 2016):** LGD 17.8% actual vs 17.5% predicted.
- **Out-of-time (2017–2021):** the model is conservative.
  - severity 32.2% predicted vs 23.5% actual;
  - LGD 5.6% predicted vs 2.0% actual.
- **Stress response:** house prices –20% at default raise LGD ×1.87.
- **EAD (D-032):** actual UPB at default / scheduled balance 12 months earlier = 1.003 (43,586 defaults).

Default year	Cure act.	Cure pred.	Severity act.	Severity pred.	LGD act.	LGD pred.
2003	65%	65%	18%	28%	4.8%	7.3%
2006	63%	63%	34%	36%	9.1%	10.4%
2008	35%	44%	43%	44%	27.6%	23.1%
2009	45%	45%	44%	41%	21.9%	20.0%
2010	48%	42%	42%	43%	19.5%	22.2%
2012	48%	51%	36%	34%	16.5%	14.2%
2015	64%	61%	37%	38%	9.5%	11.3%
2017	78%	67%	28%	34%	4.1%	7.9%
2019	83%	68%	21%	33%	1.9%	6.7%
2021	93%	79%	19%	26%	0.6%	2.8%



LGD backtest

6. CECL allowance

Method. Each loan is projected monthly to maturity, with expected loss = survival × default hazard × LGD × scheduled balance.

- **Contractual term:** adjusted for expected prepayments through the competing-risk hazard.
- **Scenarios (D-033):** three, built from the Fed's 2026 scenarios. The baseline is the Fed baseline; the downside is halfway to the severely adverse scenario; the upside is the mirror of the downside, with unemployment floored.
- **Weights:** 50% / 30% / 20% (baseline / downside / upside).
- **Forecast horizon:** a reasonable-and-supportable period of 8 quarters (D-034), then 4-quarter straight-line reversion to the historical annual loss rate of 0.145% (2000–2021, D-035).
- **Overlays (D-036):** loans that are delinquent or have a forbearance or prior-default history at the as-of date get hazard multipliers, calibrated relative to clean loans on past year-ends. Loans already in default take LGD only.

The ACL is undiscounted (a non-DCF method) on the sample portfolio (D-037).

Scenario	Weight	ACL (main)	ACL (model only)	Lifetime PD	Avg LGD	WAL (yrs)
baseline	50%	0.65%	0.27%	6.2%	5.2%	5.2
downside	30%	0.90%	0.56%	8.1%	8.1%	5.5
upside	20%	0.54%	0.17%	5.2%	3.8%	5.0
weighted	100%	**0.70%**	0.34%			

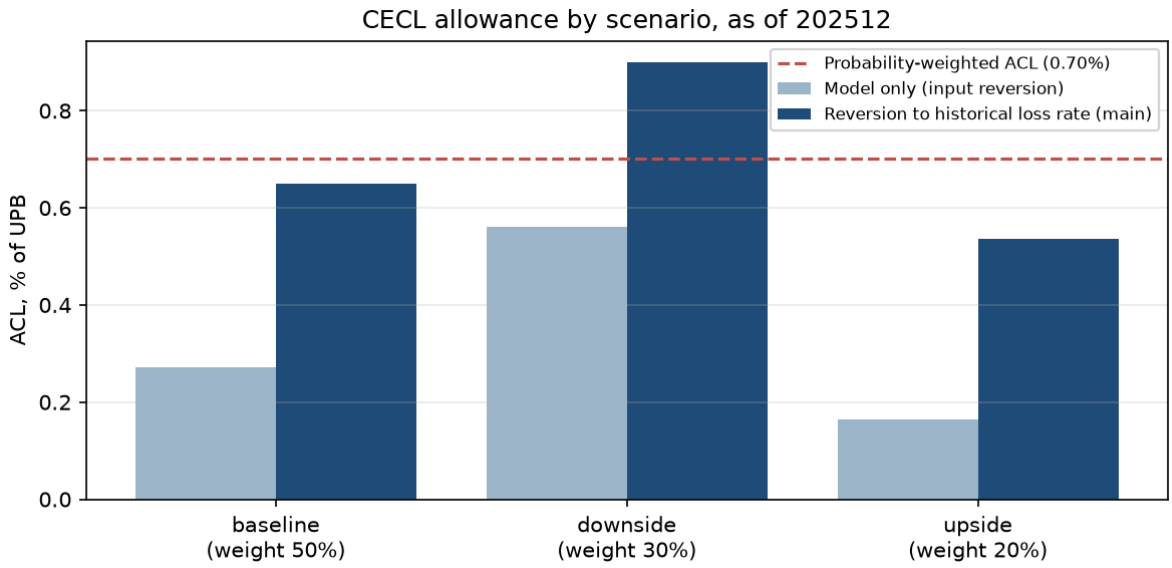
By segment and the Jan–Jun 2026 backtest. This is data never used in development. Over all segments, actual defaults were 1,858 against 2,667 predicted, and prepayments came in at 0.87× predicted.

Segment at as-of	Loans	ACL % of UPB	Jan-Jun 2026 defaults, actual	predicted
S0_clean	325,230	0.61%	322	1,054
S1_history	15,350	1.10%	196	398
S2_dpd30	3,674	3.14%	551	594
S3_dpd60	957	5.16%	524	383
S4_default	1,812	6.77%	n/a	n/a (PD = 1)
S5_forb_dq	590	4.54%	265	238

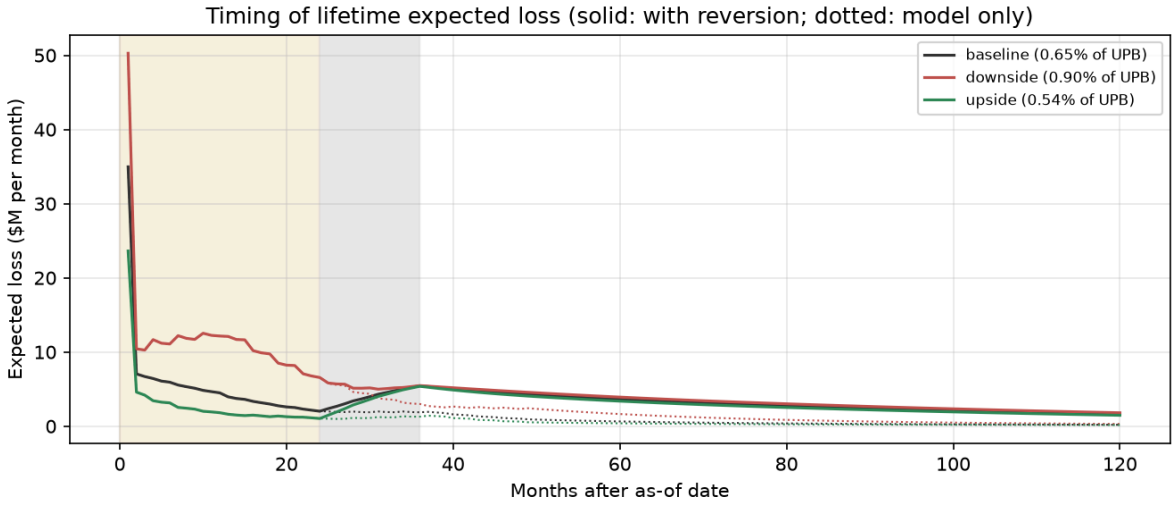
Sensitivity. The historical loss window dominates; weights, R&S length and reversion speed matter far less.

Case	ACL % of UPB	vs main
main	0.70%	+0%

Case	ACL % of UPB	vs main
model-only (input reversion)	0.34%	-52%
historical loss window 2012-2021 (0.057%/yr)	0.41%	-42%
historical loss window 2000-2006 (0.033%/yr)	0.33%	-53%
historical loss window 2008-2012 (0.501%/yr)	1.88%	+168%
weights equal	0.70%	-1%
weights base_heavy	0.69%	-2%
weights downside_heavy	0.76%	+9%
no overlays (model only for delinquent/history loans)	0.65%	-7%
R&S 4 quarters	0.72%	+2%
R&S 12 quarters	0.66%	-6%
reversion 0 quarters	0.71%	+2%
reversion 8 quarters	0.69%	-1%
downside = 100% of severely adverse	0.86%	+23%



ACL by scenario


Timing of lifetime expected loss (solid: with reversion; dotted: model only)
baseline (0.65% of UPB)
downside (0.90% of UPB)
upside (0.54% of UPB)
Expected loss ($M per month)
Months after as-of date
0
10
20
30
40
50
0
20
40
60
80
100
120

Loss timing

7. DFAST stress projection

The portfolio as of 202512 is projected over 2026Q1–2028Q1 under the Fed's published baseline and severely adverse paths (D-038). The severely adverse path peaks at 10.0% unemployment, with a national house-price change of -30% at the trough. The loss rate follows the Fed definition: cumulative loss over average balance.

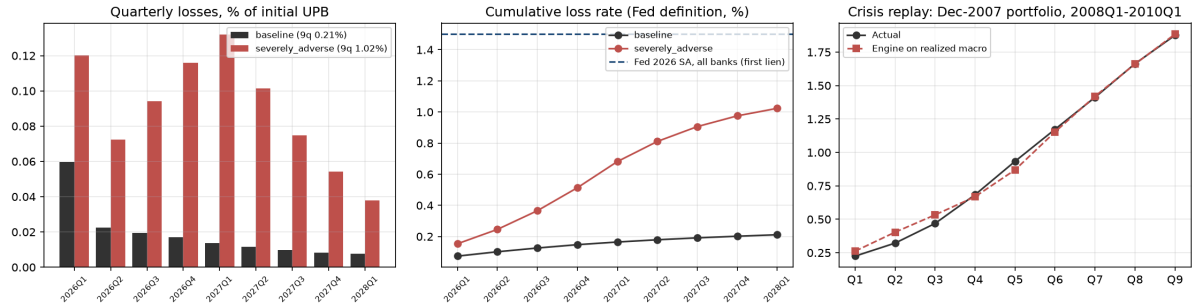
	Baseline	Severely adverse	Severely adverse, no servicing regime
9-quarter loss rate (Fed definition)	0.21%	1.02%	1.23%
LGD on defaults	6.2%	14.3%	17.2%
Cumulative default rate	2.7%	5.2%	

Crisis replay (D-039). The Dec-2007 portfolio is run on the realized 2008–2010 path: predicted 1.88% vs actual 1.88%, with LGD 21.6% vs 22.9%. The models were estimated on data that includes this period, so the replay validates aggregation and calibration under stress, not out-of-sample forecasting.

Benchmark. The severely adverse loss rate is 0.68× the Fed's all-bank first-lien figure. The differences are expected:

- this is a GSE conforming book, with MI on high-LTV loans;
- the servicing-regime assumption keeps cure rates high;
- the models are different.

Downturn LGD (D-040). Severely adverse LGD remains below the realized 2008–2010 LGD (22.1%). Switching off the servicing regime is reported as the downturn sensitivity.



DFAST losses

8. Validation evidence and traceability

The test suite has 41 of 41 tests passing (`pytest`). They cover:

- target definitions on hand-built loans;
- the layout contract;
- metric functions against independent references;
- scorecard mechanics;
- sampling correction;
- the projection engine against a closed form;
- reversion blending;
- the number guardrail and its retry loop.

Each regulatory requirement below is mapped to its implementation, tests and evidence (D-045). Assessments are the developer's own.

ID	Source	Requirement	Implementation	Tests	Evidence retrieved	Assessment
R01	SR2602a1.pdf p.6	Model testing should include out-of-sample and out-of-time testing.	Vintage-based OOT split (D-008, D-012); hazard and LGD OOT backtests (D-021, D-027)	pass	no	Met
R02	SR2602a1.pdf p.6	Testing should include a critical assessment of data quality, relevance, and inputs.	Strict schema casting, duplicate/gap/ID checks, loss identity reconciliation (D-002, D-011)	pass	no	Met
R03	SR2602a1.pdf p.6	Testing should compare alternative assumptions and methodologies.	LightGBM challenger (D-016); strict vs ex-forbearance default definitions (D-004); input vs output reversion (D-035)	pass	no	Met
R04	SR2602a1.pdf p.8	Validation of conceptual soundness assesses and documents design, key modeling choices, assumptions and data selection.	DECISIONS.md with rejected alternatives; coefficient sign checks (D-015, D-022)	pass	yes	Met

ID	Source	Requirement	Implementation	Tests	Evidence retrieved	Assessment
R05	SR2602a1.pdf p.9	Outcomes analysis compares model outputs with real-world outcomes, including back-testing.	Jan-Jun 2026 CECL backtest (D-036); 2008-2010 crisis replay (D-039)	pass	yes	Met
R06	SR2602a1.pdf p.8	Benchmarking to other models may be a practical validation assessment.	Challenger model (D-016); Fed 2026 first-lien benchmark (D-038)	pass	yes	Met
R07	SR2602a1.pdf p.9	Ongoing monitoring evaluates whether the model performs as expected as conditions change.	Score PSI and per-feature CSI (D-017); calibration by snapshot year	pass	yes	Partially met
R08	SR2602a1.pdf p.3	Generative and agentic AI are outside the guidance; governance and controls should still be determined for such tools.	LLM writes narrative only; number guardrail and citation check (D-043)	pass	yes	Met
R09	SR2602a1.pdf p.11	Adequate documentation supports continuity and tracking of recommendations and exceptions.	DECISIONS.md, metrics.json, MDD draft, this matrix	pass	no	Met
R10	acl_policy_statement_2023.pdf p.3,4	Expected credit losses are estimated over the contractual term, considering expected prepayments.	Competing-risk hazard; engine projects to maturity with prepayment (D-021)	pass	no	Met
R11	acl_policy_statement_2023.pdf p.3,4	Estimates consider past events, current conditions, and	Fed-based scenarios over an 8-quarter R&S period (D-033, D-034)	pass	yes	Met

ID	Source	Requirement	Implementation	Tests	Evidence retrieved	Assessment
		reasonable and supportable forecasts.				
R12	acl_policy_statement_2023.pdf p.4	Beyond the R&S period, revert to historical loss information using a systematic, rational method.	4-quarter straight-line output reversion; input reversion as sensitivity (D-035)	pass	yes	Met
R13	acl_policy_statement_2023.pdf p.4	Multiple economic scenarios may be considered.	Probability-weighted baseline/downside/upside (D-033)	pass	no	Met
R14	acl_policy_statement_2023.pdf p.4	Historical loss information is adjusted for differences in current conditions, e.g. delinquency.	Delinquency and forbearance overlays calibrated on past year-ends (D-036)	pass	yes	Met
R15	acl_policy_statement_2023.pdf p.5	Collateral-dependent loans are measured at fair value of collateral less costs to sell.	Loans in default take PD = 1 and the LGD model (severity includes selling and carrying costs)	pass	yes	Partially met
R16	2026_final_scenarios.pdf p.13	Supervisory projections use the published scenarios over the projection horizon.	Fed 2026 baseline and severely adverse paths, 9 quarters (D-038)	pass	no	Met

9. Limitations, assumptions and use restrictions

1. **Portfolio coverage.** GSE conforming loans with MI on high-LTV loans; not representative of jumbo, portfolio or subprime books.
2. **Forbearance.** Forbearance and disaster flags exist only from 2014, so earlier relief episodes follow the strict default definition.
3. **Through-the-cycle levels.** The hazard over-predicts clean-loan defaults in benign years (calibration-year actual/predicted 0.48), which is conservative.
4. **LGD.** Conservative in 2017–2021. It relies on a servicing-regime assumption never observed in a severe recession, and stress LGD stays below the 2008 peak unless that assumption is switched off.
5. **Reversion.** The allowance is highly sensitive to the historical loss window; that choice belongs to the allowance committee.
6. **Residual era proxy.** The hazard model still uses nominal loan size (small coefficients, clean out-of-time backtest).
7. **COVID.** Policy months are excluded from estimation. If similar policy support recurs, the model will overstate losses, which is a qualitative-adjustment matter.
8. **Sample portfolio.** Dollar amounts refer to the Freddie Mac sample; ratios are the meaningful outputs.
9. **Use restriction.** Not validated for jumbo, HELOC or commercial portfolios, or for horizons beyond the projection engine's 360-month cap.

10. Ongoing monitoring plan

Metric	Watch	Action
Score PSI (development vs current)	> 0.1	> 0.25: investigate and consider redevelopment
Out-of-time Gini	—	Drop > 0.05 vs 0.778
Actual/predicted defaults and prepayments by segment	Outside 0.8–1.25 for two consecutive quarters	Recalibrate overlay or hazard
LGD actual/predicted on resolved defaults	Outside 0.8–1.25	Review servicing regime and severity drivers

Monitoring frequency: quarterly.

11. LLM reporting layer (supporting tool, not a model input)

A local LLM (Ollama, `qwen3:8b`, embeddings `nomic-embed-text`, FAISS) supports documentation.

- **Regulatory Q&A** is retrieval-augmented over the indexed sources.
- **Draft narrative:** the LLM drafts narrative from code-formatted facts, under a number guardrail and a citation check (D-043).
- **Out of scope for SR 26-2:** generative AI falls outside the guidance (SR 26-2), so the control is the guardrail plus a logged human review.

Result of that review: drafts that passed the automated checks still contained inverted interpretations and misattributed regulatory support (`docs/mdd/review_log.md`, D-047). No LLM-drafted text is used in this document.

Metric	Value
Questions (facts verified on gold pages)	52
Retrieval hit@1 / hit@5 / MRR (dense)	58% / 88% / 0.69
Retrieval hit@1 / hit@5 / MRR (hybrid BM25, rejected)	56% / 81% / 0.66
Key-fact recall (mean / all facts)	77% / 67%
Sentence support (faithfulness proxy)	98%
Answers with numbers grounded in context	94%
Answers with valid citations	100%

Appendix A. Decision index

48 documented decisions, each with rejected alternatives in `DECISIONS.md`.

ID	Decision
D-001	Use the Freddie Mac SFLLD *sample* files, Release 48
D-002	Convert once to typed parquet with strict casting
D-003	Default event definition
D-004	Exclude 90+ DPD months during forbearance (`default_definition: ex_forbearance`)
D-005	Prepayment vs maturity
D-006	Censoring
D-007	12-month PD dataset: yearly snapshots by loan age
D-008	Out-of-time split (to be implemented in Phase 2)
D-009	National macro drivers from the Fed's own historic series
D-010	As-of date 2025-12-31 for CECL and DFAST
D-011	Data-quality findings accepted as-is
D-012	Samples: fit / in-time holdout / out-of-time
D-013	Features: only what is known at the snapshot date
D-014	Binning: monotonic WoE with optbinning
D-015	Feature selection
D-016	Challenger: LightGBM on raw features
D-017	Validation metrics
D-018	Minimum bin size exception for rare, high-risk states (bug fix)
D-019	Exclude original note rate and channel (era proxies)
D-020	Recalibrate PD levels on a recent window (scorecard ranking kept)
D-021	Discrete-time multinomial logit hazard on the loan-month panel
D-022	Only projectable drivers
D-023	Choice-based sampling with constant correction
D-024	Exclude March 2020 – December 2021 from estimation
D-025	No forbearance covariate
D-026	Forbearance history and delinquency at the forecast date are overlays
D-027	Two-part LGD on resolved defaults through 2021
D-028	Code 16 (reperforming-loan sale) counts as a cure

ID	Decision
D-029	Severity on principal, UPB at default as the denominator
D-030	Forward house-price change as an LGD driver
D-031	Servicing-regime indicator; unemployment removed from the cure model
D-032	EAD = scheduled amortization balance
D-033	Three probability-weighted scenarios built from the Fed’s 2026 scenarios
D-034	Reasonable-and-supportable period: 8 quarters
D-035	Reversion: straight-line over 4 quarters to the historical loss rate
D-036	Overlays for delinquency and history at the as-of date
D-037	Undiscounted, sample-portfolio allowance
D-041	Loan size in real terms in the LGD model (correction found in Phase 6)
D-038	Supervisory projection setup
D-039	Crisis replay as the stress-engine validation
D-040	Downturn LGD and the servicing-regime assumption
D-042	Local open-source stack only
D-043	The LLM never produces numbers: guardrail + citation check
D-044	Evaluation design
D-045	Traceability matrix
D-046	Dense retrieval kept; hybrid BM25 rejected
D-047	The guardrail is necessary but not sufficient: human review is the control
D-048	Reports rendered from metrics; reproducibility tested on a clean copy

Appendix B. Reproducibility

- **Run:** `make all` reruns phases 1–6 and the tests from raw data; `make llm` reruns the LLM layer; `make reports` renders this document.
- **Inputs:** every parameter, path, URL and the random seed (20261110) are in `config.yaml`.
- **Downloads:** public files are hashed in `data/raw/manifest_public.json`.
- **Data:** raw data is not distributed (Freddie Mac terms); see the README for download steps.

Clean-copy reproducibility test (D-048). - **Setup:** code, config and templates only, raw data linked, nothing derived. - **Run:** `make all` rebuilt phases 1–6 in 2467 seconds, with 44 tests passing. - **Comparison:** of 3,333 numeric values, 88 differ, all in Phase 2: - the LightGBM challenger is not bit-reproducible under multithreading; - two row-order-dependent scorecard diagnostics shift slightly. - **Exact matches** (maximum relative difference):

Section	Max relative difference
Data (phase 1)	0e+00
Hazard (phase 3)	9e-13
LGD/EAD (phase 4)	3e-15
CECL (phase 5)	2e-14
DFAST (phase 6)	1e-14